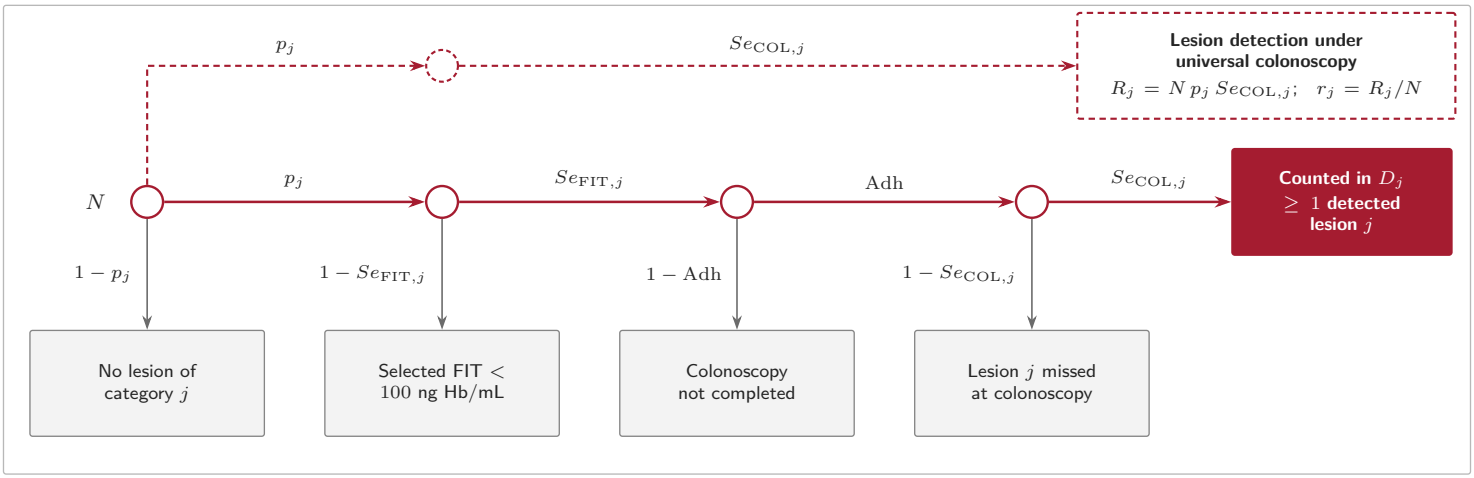

Lesion categories analysed (j): any adenoma; small (1–5 mm), medium (6–9 mm), and large ($\geq$ 10 mm) adenomas; low-risk and high-risk adenomas; and colorectal cancer.

The same scheme is applied separately to each lesion and by sex (both sexes, women, and men).

Target quantities and model-based estimators

Target quantity

$$d_j = \frac{D_j}{N} = p_j\,Se_{\mathrm{FIT},j}\,(\mathrm{Adh} \mid Se_{\mathrm{FIT},j})\,(Se_{\mathrm{COL},j} \mid \mathrm{Adh}, Se_{\mathrm{FIT},j})$$

$$p_j = \frac{d_j}{Se_{\mathrm{FIT},j}\,(\mathrm{Adh} \mid Se_{\mathrm{FIT},j})\,(Se_{\mathrm{COL},j} \mid \mathrm{Adh}, Se_{\mathrm{FIT},j})}$$

$$r_j = \frac{R_j}{N} = p_j\,Se_{\mathrm{COL},j}$$

Estimator given independence

$$d_j = \frac{D_j}{N} = p_j\,Se_{\mathrm{FIT},j}\,\mathrm{Adh}\,Se_{\mathrm{COL},j}$$

$$p_j = \frac{d_j}{Se_{\mathrm{FIT},j}\,\mathrm{Adh}\,Se_{\mathrm{COL},j}}$$

$$r_j = \frac{R_j}{N} = \frac{d_j}{Se_{\mathrm{FIT},j}\,\mathrm{Adh}}$$

Definitions of study measures

Colonoscopy detection rate (CDR_j)

Among selected-FIT-positive participants who completed colonoscopy, the proportion with at least one detected lesion of category j. It describes colonoscopy findings and does not correct for FIT sensitivity, colonoscopy adherence, or colonoscopy sensitivity.

Lesion detection under universal colonoscopy (r_j)

The estimated proportion of the target population with a lesion of category j detected under universal colonoscopy. It adjusts the single-FIT-pathway detection rate for FIT sensitivity and adherence but retains misses from imperfect colonoscopy sensitivity.

Lesion prevalence (p_j)

The estimated proportion of the target population with at least one lesion of category j after additionally accounting for colonoscopy sensitivity. This is the final prevalence estimate.

Notation *In the estimator column, hats are omitted and the same symbols denote plug-in estimates.*

L_j: participant has at least one true lesion of category j.

$p_j = \Pr(L_j)$: prevalence of lesion category j.

D_j: observed number of participants counted through the single-FIT detection pathway.

d_j: population probability of detection through that pathway; in the estimator column, $d_j = D_j/N$.

F^+: selected-FIT-positive count; C: completed-colonoscopy count among selected-FIT-positive participants.

$Se_{\mathrm{FIT},j}$: sensitivity of the selected FIT at the 100 ng Hb/mL threshold for lesion category j.

$\mathrm{Adh} = C/F^+$: colonoscopy-adherence probability among participants whose selected FIT is $\geq$ 100 ng Hb/mL.

$Se_{\mathrm{COL},j}$: colonoscopy sensitivity for lesion category j among participants completing colonoscopy.

r_j: lesion detection under universal colonoscopy; $R_j = Nr_j$: corresponding expected count.

$\mathrm{CDR}_j = D_j/C$: observed colonoscopy detection rate.

Conditional-independence assumptions

1. Adherence: $(\mathrm{Adh} \mid Se_{\mathrm{FIT},j}) = \mathrm{Adh}$

Colonoscopy adherence is assumed not to vary with FIT sensitivity.

2. Colonoscopy sensitivity: $(Se_{\mathrm{COL},j} \mid \mathrm{Adh}, Se_{\mathrm{FIT},j}) = Se_{\mathrm{COL},j}$

Colonoscopy sensitivity is assumed not to vary with adherence or FIT sensitivity.

Colonoscopy plus histological ascertainment is assumed to have effectively perfect specificity, so no false-positive term is added. The model is applied separately to each lesion category and sex.